

# **Investigations of Solutions to the Different Pedigree and Phylogeny Rates of Evolution in Human Mitochondrial Genomes**

## **The Rationale for the Study**

The human mitochondria DNA functions independently from our nuclear or cell-free DNA, operating with their own machinery for replication and repair pathways, as well as having different inheritance patterns to our nuclear genome (1). Studying the inheritance patterns of the mitochondrial DNA allows us to grow our understanding of molecular evolution and explore how we as a species will and have changed. “Mutation rates” is a term used in the study of molecular evolution to describe the speed at which certain molecules change due to random variation. Most commonly, we use the term “mutation rates” when referring to the number of changes that occur to the nucleic acid bases that make up our genome over a period of time. This term is not to be confused with “substitution rates”, which refers to the implantation of genetic variation in a population (2). Mutations that stick and get inherited over generations differ from the single nucleotide variation we observe between a mother and their child for example, which is what the term “mutation rate” directly references.

In population genetics, we can use the analysis of mutation rates and substitution rates to estimate divergence of species and population splits by dating back current genomic data in comparison to data of previous primate taxa such as chimpanzees (3). Analysis of mitochondrial DNA sequence data can also aid in this task without the burden of whole-genome sequencing. There are two main methods of identifying the mutation and substitution rate in human mitochondrial DNA: One being pedigree analysis and the other being phylogenetic analysis, short-term and long-term evolutionary genetic analysis respectively (4). In both these studies, human pedigree rates nearly differ by 1 to 2 orders of magnitude higher than rates collected in

species phylogenetic studies (2-4). The average pedigree analysis depicts a rate of 0.95 per base pair per million years (bp/Myr) while phylogenetic studies found rates between 0.12 and 0.38/bp/Myr (3).

The reason behind this discrepancy has been a high-interest area of study with multiple hypotheses suggested explaining why we observe this change (4). Studies by Simon Ho have suggested that selection within the mitochondrial demographic, sequence error due to low variation rates or mutational “hot spots” on the mitochondrial genome could be areas of research to resolve this problem (2). These areas of study collectively lead to the idea of “Time dependency” of mitochondrial mutation rate estimates that serves currently as the most compelling theory explaining the differences in mutation or substitution rates (3). This idea incorporates the theories of “hot spots” and selection of mutations, explaining an idea that rates of human mitochondrial DNA change based on the calibration times used to adjust for different timescales (2,3). A rate of decay formula has been postulated that emulates an exponential decay curve that plateaus based around a calibration point (2). This calibration point serves as a theoretical “barrier” that divides long-term evolutionary rates from short-term evolutionary rates. These calibration points are determined through archaeological evidence for phylogenetic analysis, usually the estimate of when a population split occurs (2,3). For example, a population bottleneck followed by a spike in population growth could cause different rates of evolution, as the boom of population size would increase the diversity of a restricted population, opening the availability for variation (3).

The point at which this event occurs could be the time at which the interspecies substitution rates shifts to the intraspecies mutation rates observed in pedigree analysis (2,3). The

aim of this study is to provide evidence to validate or disprove the theory of timescales of evolution for human mitochondrial genomes.

This project aims to reflect and apply the bioinformatics skills and knowledge gained through the major. Bioinformatic tools and techniques have allowed for the rapid collection of data from databanks through data mining processes. This study uses data taken from phylogenetic and pedigree analysis and through our methods we will demonstrate the ability to search and collect usable data from databases and previous mitochondria genome studies. Fundamental knowledge of how to create and understand pedigrees and phylogenies from genome sequence data will also be reflected in this study. In bioinformatics research, the use of genome sequence data is very common in most if not all studies and the application of taking sequence data and identifying variants between individuals within a pedigree or a phylogeny will be displayed in our methods. In the methods, we outline the use of SimTree and WFES2 programs to run simulations of our mutation and substitution rates. The use of computational software is a key aspect of bioinformatics research, which aims to advance the field of biology using computational tools such as software that could not be explored previously. Our study will reflect these techniques and will further our understanding of mitochondrial DNA inheritance patterns and develop a new understanding of intraspecies and interspecies mitochondrial population genetics. Using the bioinformatics tools and methods in our study we want to resolve the dispute outlined in previous literature about the mutation and substitution rates in the mitochondrial genome which would not be previously possible.

## **Literature Review**

The original study of the pedigree mutation rate and phylogeny substitution rate differences were brought up by Sigurğardóttir to discuss possible explanations for the observed differences in projects involving mitochondrial DNA (4). Sigurğardóttir observed 285 Icelandic samples from data taken from 25 pedigrees all linked to a single female ancestor, totalling 745 observable mitochondrial DNA transmission events (4). For every sample used in their analysis, they also sampled a maternally closely related individual to reduce the chance of database errors being undetected and that germline and somatic mutations are differentiable for their analysis (4). In this study, they were able to rule out database, handling and pedigree labelling errors which would lead to an overestimation of pedigree-based mutation rates (4). The prominent idea from Sigurğardóttir's study of mitochondrial mutation rates is the idea of variability or heterogeneity of mutation rates in different sites on the mitochondrial genome (4). These "hot spot" regions on the mitochondrial genome have higher mutation rates due to looser bound chromatin, and these regions can cause an underestimation of substitution rate in long-term phylogenetic studies in which double mutations or backwards mutations can be undetected (4,5). Sigurğardóttir concluded that it was just not enough to solely recalibrate mutation rates based on time and not consider mutation rate heterogeneity in models of mitochondria mutation estimates over time (4).

Ho continued these ideas with the understanding that the rate disparity is evidence for purifying selection of the mitochondrial genome that voids the neutral theory of evolution (2). Ho observed avian and primate phylogenies with three groups of mitochondria data: 7 alignments of avian protein-coding genes, 4 alignments of primate protein-coding genes, and 7 alignments of d-loop sequences (2). This study investigated the short-term mutation rates and the

long-term substitution rates under the single term “rate of change” (2). They proposed the use of different timescale calibration events and tested the rate of change from different events to solve the differential “rate of change” problem, using six different calibration points ranging from 12 million years ago to 1.4 million years ago (2). What was found was a direct correlation between calibration point and estimated “rate of change” (2). Calibration points of greater than 1-2 million years for both avian and primate mitochondria data show much higher “rates of change” than those calibrated with older population events (2). The decrease in the “rate of change” as older calibration points are used shows that some selection is present, which voids the neutral theory of evolution (2). This theory claims that most mutational evolution is due to random chance rather than a type of selection (2). With this variation in the “rate of change” dependent on time, there is some selection (most likely purifying selection that attempts to remove harmful, deleterious variants from a population) that is choosing to remove some variants and keep only a few over long periods of time (2). Ho also concluded that mutation “hot spots” that may have mutations hidden within them due to their high mutation rate (a mutation that mutates again will look like a single mutation) only account for a small fraction of the possible overestimation of short-term “rates of change” and is not enough to explain the calibration date and “rate of change” estimation correlation (2).

Using the timescale-dependent theory of mitochondrial DNA evolution, studies have been done exploring its use in solely human populations. The theory of hidden mutations due to multiple mutations in “hot spots” has been shown to reflect the inaccuracy of substitution models on mitochondrial evolution in longer-timescale phylogenetic studies such as human-chimpanzee studies (3). This reasoning, however, is not applicable to shorter-timescale studies with calibration points of 10 to 50 thousand years where the time for multiple hits to occur can not

justify the high mutation rates from intraspecies human pedigree analysis (2). The Henn study observed specifically the hypervariable region and coding regions of the mitochondria in the intraspecies human pedigree analysis as well as compared to a human-chimpanzee study like ones done by Tamura and Nei (3,6). Henn proposed a method of allowing substitution models to depict multiple hits (3). Using the K2P model of substitution, an “expected” number of substitutions over a set time (less than 6 million years) was obtained through simulating two diverging species (3). This was done by calculating a pedigree-based spontaneous/random mutation rate that was used to simulate nucleic substitution (3). After this rate was calibrated against a known timescale calibration point, which calibrates the mutation rate reflect conditions after multiple mutations have occurred (this was simulated through the random spontaneous mutations) (3). The K2P model was able to also account for “hot spots” in the mitochondrial genome and had different mutation rates for different sites (3).

This new calibrated estimate for substitution rate was able to match the observed substitution rate found from human-chimpanzee studies and simulate them (3). For within-human studies, this was not the case as the rate of substitution was far too slow to match the observed high mutation rate for calibration points between 10 and 50 thousand years (3). Henn hypothesized that purifying selection and population bottleneck events could explain the high rates in short-term pedigree analyses (3). Populations with reoccurring bottlenecks have been shown to have reduced mutation rates in comparison to populations with stable population sizes (3). Also, populations with high selection pressure have been shown to also decrease mutation rate estimates due to the removal of certain variants (3). Both these events have little to no impact on recent pedigree samples compared to older ones, which can explain the lower mutation rates in older calibrated data (3). Henn also found that calibration points after known

events of increased population growth such as after the Younger Dryas cycle would reduce the effect of past serial bottlenecks (during the Younger Dryas cycle) and these events could determine the calibration points which should be targeted to effectively define the timescale dependent mutation and substitution rate estimates (3).

### **Objective, Research Questions, Aims and Hypothesis of the Study**

The main objective of our study is to evaluate potential reasons for the differential rates between pedigree studies and phylogenetic studies of human mitochondrial DNA. With this understanding, we want to develop an understanding of how to model mitochondrial DNA variation in future population genetic studies accurately over both long-term and short-term calibrated data. Ultimately, we want to be able to find the most likely calibration point and depict accurately sites with different mutation rates that will illustrate what is seen in our mitochondrial evolutionary genomic data and previous studies.

Specifically, this study aims to target three main research questions. What is the mutation rate for variants in the region of the human mitochondria found in our pedigree data? What is the substitution rate for variants in the region of the human mitochondria found in our phylogenetic data of a recent and distant ancestor? Finally, we will ask, Does the timescales of evolution hypothesis explain accurately how mitochondrial mutations are substituted in a population and is the timescales of evolution hypothesis able to model in simulations of where calibration points exist? Potentially we could ask the simulation, what does the most realistic calibration point for the data provided?

There are 3 main aims of this study. The first being to infer mitochondrial mutation rate in humans from mother-child duo and pedigree data. We will search for a database containing mother-child mitochondria sequence data or obtain data from a previous study with mitochondria

pedigree data. We will then determine a per-site mutation rate per generation that can be extrapolated to reflect longer timescales. The second aim is to infer mitochondrial substitution rate at neutral sites in humans to distance and close ancestors. We want to be able to replicate the conflict from previous studies on further timescales having lower substitution rates to closer timescales. For this, we would obtain human and ancestral mitochondrial sequence data or phylogenetic data from a previous study and determine a substitution rate at various time scales that reflect the pedigree/phylogeny conflict in previous literature. Aim three looks to verify the timescales of evolution theory using simulations in SimTree and WFES2 for pedigree and phylogenetic data. We will achieve this by modelling the timescales of molecular evolution theory and implement the mutational “hot spots” proposed by Ho and Henn to our dataset and observing at what conditions can we reflect what is seen in our collected dataset (2,3).

The time scale of evolution will be an accurate model for mitochondrial DNA evolution but will be demographic dependent as different demographics may have experienced population growth or shunting events that reflect the variable rates of evolution seen in previous studies. The implementation of site-dependent rates of evolution will also increase the accuracy of our model.

### **Sex and Gender Consideration**

Here in this study, we plan to use anonymous mitochondria genome data from different demographic groups. In our study, we acknowledge the inheritance pattern of mitochondrial DNA being majority maternal. However, the evolution of mitochondrial DNA does not have different rates based on the sex of the individual. Gender does not have an impact on mitochondrial DNA evolution either. Sex and gender impacts have been considered in this study, and as the project is being completed any notice of Sex or Gender-based results will be mentioned and the methods will be adapted for such events to take Sex and Gender into account.

## **Research Approach and Methods**

We will obtain data from either a database containing mitochondrial sequence information of families with a common ancestor or from a study done with mitochondrial pedigree or phylogenetic analysis. The latter will most likely be the case as this data can be verified back to the original study and any database errors would be already mentioned. For data taken prior to any analysis (sequence database data), we will align sequences and develop the most likely phylogeny or pedigree-based on sequence similarity. From this, we can begin the analysis of mutation and substitution rates. We will locate a common diverging point in both analyses which will serve as a starting time for our rate calculation.

For phylogenetic analyses, we will locate common ancestors with known archaeological data that suggests a time of divergence which would allow for the calculation of a substitution rate between each sample. These rates can then be plotted and compared to previous studies of human mitochondrial evolution to evaluate the rate of mutational decay accuracy. For pedigree analyses, we will use our created pedigrees and mitochondrial sequence difference to determine a per generation variation rate that would allow us to extrapolate to a larger timescale and verify the long-term versus short-term analysis differences.

From here, we will take the results from these two analyses and attempt to model and simulate short- and long-term mitochondrial mutation for hypothetical randomly generated sequences. After which we will compare the accuracy of the simulations to what is observed from the first analyses and possibly previous studies. To quantitatively determine accuracy, a statistical difference test will be done between the rate of evolution curves from our simulation and previous study and our own analyses curves. A cut-off point for accuracy will be determined based on the current literature. For our simulation, we will use SimTree, an R package that

simulates phylogenetic trees. We will also use WFES2, a Wright-Fisher solver that can compute biologically realistic sizes of a simulated population.

### **Knowledge Translation and Possible Benefits of the Study**

The results obtained will lead to a greater understanding of molecular evolution and provide evidence of the pedigree versus phylogeny evolution rate differences with the most promising explanations being further explored. Individuals interested in our mitochondrial genomic studies will be able to view the finding from this study with the provided methods and datasets with full anonymity of the participants sampled in the study. Data taken from previous studies will be referenced. The results of this study can allow for knowledge gaps to be filled. In previous literature studies of mitochondrial DNA evolution, it was observed that a lot of the conclusions could not account for some errors due to lack of data, or not robust enough software analysis techniques. In this study, we aim to loosen some of those restrictions from the past. Our objective is to resolve the controversy and determine a modern frontrunner for the most likely explanation for the mitochondrial genome mutation rate controversy.

## **References:**

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- 2) Ho SY, Phillips MJ, Cooper A, Drummond AJ. Time dependency of molecular rate estimates and systematic overestimation of recent divergence times. *Molecular Biology and Evolution*. 2005;22(7):1561–8.
- 3) Henn BM, Gignoux CR, Feldman MW, Mountain JL. Characterizing the time dependency of human mitochondrial DNA mutation rate estimates. *Molecular Biology and Evolution*. 2008;26(1):217–30.
- 4) Sigurðardóttir S, Helgason A, Gulcher JR, Stefansson K, Donnelly P. The mutation rate in the human mtdna control region. *The American Journal of Human Genetics*. 2000;66(5):1599–609.
- 5) Innan H, Nordborg M. Recombination or mutational hot spots in human mtdna? *Molecular Biology and Evolution*. 2002;19(7):1122–7.
- 6) Tamura K, Nei M. Estimation of the number of nucleotide substitutions in the control region of mitochondrial DNA in humans and chimpanzees. *Molecular Biology and Evolution*. 1993;

## **Project timeline**

### **OCT 29TH RESEARCH EXCHANGE – DEADLINE TO BEGIN PHYLOGENY AND PEDIGREE DATA COLLECTION**

- Obtain dataset from database or previous study of mitochondria genomes.

### **NOV 29th – DEADLINE TO FINISH PEDIGREE OR PHYLOGENY**

- Obtain Rate 1 (mutation or substitution rate).

### **DECEMBER 29th – DEADLINE TO FINISH EITHER PEDIGREE OR PHYLOGENY**

- Obtain Rate 2 (mutation or substitution rate not obtained above).

### **Jan 14th PROGRESS REPORT – DEADLINE TO BEGIN SIMULATIONS (AIM 3)**

- Start using Simtree and WFES2 to determine best Timescales of evolution decay rate for human mitochondrial DNA.

### **Jan 21st RESEARCH IN PROGRESS 1-1**

### **FEB 4th - DEADLINE TO START WRITING THESIS**

- Start background or Result section with results found so far.

### **FEB 11th RESEARCH IN PROGRESS PRESENTATION**

### **FEB 20th FINISH GETTING DATA – DEADLINE TO FINISH SIMULATIONS**

- Get enough data from simulations to determine the accuracy of different timescales.

### **Mar 4th THESIS DRAFT**

- Give to TA and PI

### **Mar 25th THESIS DUE**

### **Mar 29th – DEADLINE TO START ORAL PRESENTATION PREPARATION**

### **April 8th – ORAL PRESENTATIONS**